Crossing Symmetry: Compton → Pair Annihilation

How to Turn $\gamma e^- \to \gamma e^-$ into $e^+ e^- \to \gamma\gamma$

Griffiths Problems 7.30/7.41 | Lancaster & Blundell §40.4 | Thomson §6.5.6

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1 The Big Idea

Crossing Symmetry in One Sentence

An incoming particle with 4-momentum p can be "crossed" to an outgoing antiparticle with 4-momentum $-p$, and the amplitude is *the same function* of the momenta.

This means: if you know $|\mathcal{M}|^2$ for one process, you can get $|\mathcal{M}|^2$ for a whole family of related processes by *substituting momenta*. No new calculation required.

Why It Works: Feynman's Antiparticle Interpretation

Feynman's key insight: *an antiparticle traveling forward in time is equivalent to the particle traveling backward in time.* In a Feynman diagram, reversing an external leg from "incoming" to "outgoing" (and flipping particle $\leftrightarrow$ antiparticle) is achieved by simply negating its 4-momentum: $p \to -p$.

The matrix element $\mathcal{M}$ is an *analytic function* of the external momenta. Crossing symmetry is the statement that this same analytic function, evaluated at different values of the momenta (some with signs flipped), describes physically different processes. Lancaster & Blundell (Eq. 40.17):

$$\mathcal{M}\big(\phi(p) + \ldots \to \ldots\big) = \mathcal{M}\big(\ldots \to \ldots + \bar{\phi}(k)\big), \qquad p = -k$$

2 The Two Processes: Compton and Pair Annihilation

2.1 Process 1: Compton Scattering

$$\boxed{e^-(p) + \gamma(k) \;\longrightarrow\; e^-(p') + \gamma(k')}$$

(a) s-channel

$e^-(p),\ \gamma(k) \;\to\; [e^-_{\text{virt}}, q = p + k] \;\to\; e^-(p'),\ \gamma(k')$

Propagator: $\dfrac{1}{(p+k)^2 - m^2} = \dfrac{1}{s_C - m^2}$

(b) u-channel

$e^-(p) \;\to\; [e^-_{\text{virt}}, q = p - k'] \;\to\; e^-(p'),\ \gamma(k');\ \gamma(k)$

Propagator: $\dfrac{1}{(p-k')^2 - m^2} = \dfrac{1}{u_C - m^2}$

Mandelstam variables (Compton):

$$s_C = (p+k)^2 \qquad t_C = (p-p')^2 \qquad u_C = (p-k')^2 \tag{1}$$

Propagator denominators: s-channel $\to s_C - m^2$, $\quad u$-channel $\to u_C - m^2$.

2.2 Process 2: Pair Annihilation

$$\boxed{e^-(p_1) + e^+(p_2) \;\longrightarrow\; \gamma(k_1) + \gamma(k_2)}$$

(a) t-channel

$e^-(p_1) \;\to\; [e^-_{\text{virt}}, q = p_1 - k_1] \;\to\; \gamma(k_1)$
$e^+(p_2) \;\to\; \gamma(k_2)$

Propagator: $\dfrac{1}{(p_1 - k_1)^2 - m^2} = \dfrac{1}{t_A - m^2}$

(b) u-channel

$e^-(p_1) \;\to\; [e^-_{\text{virt}}, q = p_1 - k_2] \;\to\; \gamma(k_2)$
$e^+(p_2) \;\to\; \gamma(k_1)$

Propagator: $\dfrac{1}{(p_1 - k_2)^2 - m^2} = \dfrac{1}{u_A - m^2}$

Mandelstam variables (Annihilation):

$$s_A = (p_1 + p_2)^2 \qquad t_A = (p_1 - k_1)^2 \qquad u_A = (p_1 - k_2)^2 \tag{2}$$

Propagator denominators: t-channel $\to t_A - m^2$, $\quad u$-channel $\to u_A - m^2$.

Key Observation: Diagram Topology Changes!

Notice something critical: Compton has s- and u-channel diagrams (electron propagator), but pair annihilation has t- and u-channel diagrams. The crossing has turned the *s-channel into the t-channel*! This makes physical sense:

- In Compton, the s-channel has the electron absorbing the photon $\to$ virtual electron $\to$ emitting a photon. The "time ordering" flows left-to-right.
- In pair annihilation, there is *no* s-channel diagram at tree level (you can't have $e^+e^- \to$ single virtual electron $\to \gamma\gamma$; the vertex structure doesn't allow it). Instead, the electron propagates in the t-channel, exchanged between the two photon emission vertices.

3 The Crossing: Step by Step

The Crossing Prescription

To go from Compton $e^-(p) + \gamma(k) \to e^-(p') + \gamma(k')$ to pair annihilation $e^-(p_1) + e^+(p_2) \to \gamma(k_1) + \gamma(k_2)$, we perform two crossings:

Cross 1: Take the *incoming photon* $\gamma(k)$ and cross it to an *outgoing photon* $\gamma(k_1)$:

$$k \longrightarrow -k_1 \qquad \text{(incoming} \to \text{outgoing: negate momentum)}$$

Since the photon is its own antiparticle, no particle identity change is needed.

Cross 2: Take the *outgoing electron* $e^-(p')$ and cross it to an *incoming positron* $e^+(p_2)$:

$$p' \longrightarrow -p_2 \qquad \text{(outgoing particle} \to \text{incoming antiparticle: negate momentum)}$$

The remaining legs are left as is:

$$p \longrightarrow p_1, \qquad k' \longrightarrow k_2$$

Summary of Momentum Replacements

$$\boxed{p \to p_1, \qquad k \to -k_1, \qquad p' \to -p_2, \qquad k' \to k_2} \tag{3}$$

4 What Happens to the Mandelstam Variables

This is the punchline. Apply the substitutions (3) to the *Compton* Mandelstam variables:

$$s_C = (p+k)^2 \longrightarrow (p_1 - k_1)^2 = t_A \tag{4}$$

$$t_C = (p-p')^2 \longrightarrow (p_1 + p_2)^2 = s_A \tag{5}$$

$$u_C = (p-k')^2 \longrightarrow (p_1 - k_2)^2 = u_A \tag{6}$$

Mandelstam Variable Mapping

$$\boxed{s_C \longrightarrow t_A, \qquad t_C \longrightarrow s_A, \qquad u_C \longrightarrow u_A} \tag{7}$$

In words: **the s and t variables swap**, while u **stays the same**.

Why $s \leftrightarrow t$ Makes Physical Sense

Consider what each variable *means*:

- s = (total incoming)^2: In Compton, this is $s_C = (p+k)^2$, the invariant mass of the $e^-\gamma$ system. In annihilation, the total incoming momentum is $(p_1+p_2)^2 = s_A$. But $(p+k)^2$ *maps to* $(p_1 - k_1)^2 = t_A$, not s_A!
- This is because the photon k got crossed to the *other side* (outgoing), so $p+k$ became $p_1 - k_1$, which is a *momentum transfer*, not a total energy.

The Compton s-channel propagator (virtual electron with momentum $p+k$) becomes the annihilation t-channel propagator (virtual electron with momentum $p_1 - k_1$). This is exactly the diagram topology change we noted above!

5 Applying the Crossing to $\overline{|\mathcal{M}|^2}$

5.1 Starting Point: Compton Result

From the full Compton calculation (Griffiths 7.30 + H&M $-g_{\mu\nu}$ trick), in the massless limit:

$$\overline{|\mathcal{M}|^2}_{\text{Compton}} = -2e^4\left(\frac{u_C}{s_C} + \frac{s_C}{u_C}\right) \tag{8}$$

5.2 Apply the Crossing Map

Substitute $s_C \to t_A$ and $u_C \to u_A$ (and $t_C \to s_A$, though it doesn't appear explicitly):

$$\overline{|\mathcal{M}|^2}_{\text{Compton}}\Big|_{\substack{s_C \to t_A \\ u_C \to u_A}} = -2e^4\left(\frac{u_A}{t_A} + \frac{t_A}{u_A}\right) \tag{9}$$

5.3 Correction Factor: Spin Averaging

There is one subtlety: the $\frac{1}{4}$ averaging factor differs between the two processes.

- **Compton:** Average over 2 electron spins $\times$ 2 photon polarizations $= \frac{1}{4}$.
- **Pair annihilation:** Average over 2 electron spins $\times$ 2 positron spins $= \frac{1}{4}$.

In this case, both factors are $\frac{1}{4}$, so the crossing goes through directly with *no additional correction*. (This is lucky; in general, the statistical prefactors can differ.)

Pair Annihilation Result via Crossing (Griffiths Problem 7.41)

$$\boxed{\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma} = -2e^4\left(\frac{u}{t}+\frac{t}{u}\right) = -2e^4\,\frac{t^2+u^2}{tu}} \tag{10}$$

where the subscripts on the Mandelstam variables are now dropped (they refer to the annihilation process).

Physical region check: For $e^+e^-\to\gamma\gamma$, we have $s>0$, $t\le 0$, $u\le 0$, so:

$$-\frac{u}{t}>0, \qquad -\frac{t}{u}>0 \qquad \Longrightarrow \qquad \overline{|\mathcal{M}|^2}>0 \quad \checkmark$$

6 Verification: Direct Calculation of Pair Annihilation

Let us verify this by examining the pair annihilation amplitude directly and checking that the trace structure matches.

Pair Annihilation Amplitudes (Griffiths Eqs. 7.133–7.134)

For $e^-(p_1)+e^+(p_2)\to\gamma(k_1)+\gamma(k_2)$:
t-channel (electron exchange):

$$\mathcal{M}_t = \frac{-e^2}{t-m^2}\,\bar{v}(p_2)\,\not{\epsilon}_2\,(\not{p}_1-\not{k}_1+m)\,\not{\epsilon}_1^*\,u(p_1) \tag{11}$$

u-channel (electron exchange with photons swapped):

$$\mathcal{M}_u = \frac{-e^2}{u-m^2}\,\bar{v}(p_2)\,\not{\epsilon}_1^*\,(\not{p}_1-\not{k}_2+m)\,\not{\epsilon}_2\,u(p_1) \tag{12}$$

Note: there is a relative minus sign between $\mathcal{M}_t$ and $\mathcal{M}_u$ from Fermi statistics (identical photons in final state), already absorbed into the sign convention.

Comparing with the Compton amplitude:
The Compton s-channel amplitude was (Eq. **??**):

$$\mathcal{M}_s^{\text{Comp}} = \frac{-e^2}{s-m^2}\,\bar{u}(p')\,\not{\epsilon}'^*(\not{p}+\not{k}+m)\not{\epsilon}\,u(p)$$

Under the crossing $p\to p_1$, $k\to -k_1$, $p'\to -p_2$:

- $\bar{u}(p')\to\bar{u}(-p_2)\sim\bar{v}(p_2)$ (outgoing particle $\to$ incoming antiparticle)
- $\not{p}+\not{k}\to\not{p}_1-\not{k}_1$ (the internal propagator momentum)
- $s-m^2=(p+k)^2-m^2\to(p_1-k_1)^2-m^2=t-m^2$

So the Compton s-channel amplitude *becomes* the annihilation t-channel amplitude, exactly as expected. The same happens for the u-channel ($u_C\to u_A$).

The $\bar{u} \to \bar{v}$ Replacement and Signs

When we cross an outgoing electron to an incoming positron, the spinor replacement is $u(p') \to v(p_2)$ with $p' = -p_2$. In the completeness relation, this changes $\sum u\bar{u} = \not{p}' + m$ to $\sum v\bar{v} = \not{p}_2 - m$. But since $p' = -p_2$, we have $\not{p}' + m = -\not{p}_2 + m = -(\not{p}_2 - m)$. The sign flip from $+m \to -m$ is *automatically* accounted for by the momentum negation $p' \to -p_2$. This is why crossing "just works" at the level of $|\mathcal{M}|^2$ expressed in Mandelstam variables—the Mandelstam substitution encodes both the momentum change and the particle $\to$ antiparticle swap.

7 The Invariant Cross Section $d\sigma/dt$

Lorentz-Invariant Cross Section Formula (Thomson Eq. 3.37)

$$\frac{d\sigma}{dt} = \frac{\overline{|\mathcal{M}|^2}}{64\pi\, s\, p_i^{*2}}$$

For $e^+e^- \to \gamma\gamma$ with $m = 0$: $p_i^{*2} = s/4$, so:

$$\frac{d\sigma}{dt} = \frac{\overline{|\mathcal{M}|^2}}{16\pi\, s^2}$$

Substituting the pair annihilation result (10):

$d\sigma/dt$ for Pair Annihilation ($m = 0$)

$$\boxed{\left.\frac{d\sigma}{dt}\right|_{e^+e^-\to\gamma\gamma} = -\frac{2\pi\alpha^2}{s^2}\left(\frac{t}{u} + \frac{u}{t}\right) = -\frac{2\pi\alpha^2(t^2+u^2)}{s^2\, tu}} \tag{13}$$

Using $u = -s - t$:

$$\frac{d\sigma}{dt} = \frac{2\pi\alpha^2\left[t^2 + (s+t)^2\right]}{s^2\, t\,(s+t)} \tag{14}$$

For comparison, the Compton result was:

$$\left.\frac{d\sigma}{dt}\right|_{\text{Compton}} = -\frac{2\pi\alpha^2(s^2+u^2)}{s^3\, u}$$

Notice the structural similarity: same overall coefficient $-2\pi\alpha^2$, but with $s \leftrightarrow t$ swapped in the numerator and denominator.

8 The Full Crossing Web: All Related Processes

The Compton–Annihilation–Pair Production Family

All three processes are related by crossing symmetry, and their $|\mathcal{M}|^2$ are obtained from the *same analytic function* $F(s,t,u) = -2e^4\dfrac{a^2+b^2}{ab}$ with different assignments of which Mandelstam variable plays which role:

Process	Channels	$\overline{\lvert\mathcal{M}\rvert^2}$	Obtained by
$e^- + \gamma \to e^- + \gamma$ (Compton)	$s + u$	$-2e^4\left(\frac{u}{s} + \frac{s}{u}\right)$	direct calculation
$e^+e^- \to \gamma\gamma$ (Pair annihilation)	$t + u$	$-2e^4\left(\frac{u}{t} + \frac{t}{u}\right)$	$s_C \to t$, $t_C \to s$, $u_C \to u$
$\gamma\gamma \to e^+e^-$ (Pair production)	$t + u$	$-2e^4\left(\frac{u}{t} + \frac{t}{u}\right)$	time reversal of annihilation

The pattern: each result involves the **two Mandelstam variables that label the propagator channels**. Compton has s- and u-channel propagators $\Rightarrow$ s and u appear. Annihilation has t- and u-channel propagators $\Rightarrow$ t and u appear.

9 Summary: The Crossing Recipe

Complete Crossing Symmetry Recipe

1. **Identify the crossing.** Decide which legs to flip:
   - Incoming particle $\to$ outgoing antiparticle: $p \to -p$
   - Outgoing particle $\to$ incoming antiparticle: $p \to -p$
2. **Map the momenta.** For Compton $\to$ Annihilation:
$$p \to p_1, \quad k \to -k_1, \quad p' \to -p_2, \quad k' \to k_2$$
3. **Map the Mandelstam variables.** Substitute the momentum map into the definitions of s, t, u:
$$s_C \to t_A, \quad t_C \to s_A, \quad u_C \to u_A$$
4. **Substitute into $\overline{\lvert\mathcal{M}\rvert^2}$.** Replace every s, t, u in the Compton result:
$$-2e^4\left(\frac{u}{s} + \frac{s}{u}\right) \xrightarrow{s\to t,\, u\to u} -2e^4\left(\frac{u}{t} + \frac{t}{u}\right)$$
5. **Check averaging factors.** Both processes have $\frac{1}{4}$ $\Rightarrow$ no correction needed.
6. **Check physical region.** Verify $\overline{\lvert\mathcal{M}\rvert^2} > 0$ in the new physical region.
7. **Compute $d\sigma/dt$.** Plug into $\dfrac{d\sigma}{dt} = \dfrac{\overline{\lvert\mathcal{M}\rvert^2}}{16\pi s^2}$ (for massless particles).

The Deep Lesson

Crossing symmetry is not just a computational trick—it reflects a profound property of quantum field theory. The *same* Feynman diagram, analytically continued to different kinematic regions, describes physically different processes. Compton scattering and pair annihilation look

completely different in the lab, but at the level of the S-matrix they are the same analytic function evaluated at different points.
As Griffiths puts it (Section 1.4): *"The power and beauty of crossing symmetry can scarcely be exaggerated. It tells us, for instance, that Compton scattering $\gamma + e^- \to \gamma + e^-$ is 'really' the same process as pair annihilation $e^+ + e^- \to \gamma + \gamma$, although in the laboratory they are completely different phenomena."*